SUPPLEMENTARY MATERIAL S1

Candidate readings of the Xiongnu record

Forty readings of Chinese-transcribed Xiongnu names and titles as Turkic words, with the historical parallels that suggested each one. Each reading priced against the measurements in the paper. Supplied for the record; not offered as results.

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Supplement to *Reading steppe names: what a transcription channel can and cannot recover from the Chinese record of the Xiongnu*

S1 · author_proposals.csv · compiled 2026

S1.1 What this supplement is

Why the file is worth having, given that none of it is a finding. A reading of a Xiongnu name has been offered many times over a century, and offered as a resemblance: this Chinese spelling looks like that Turkic word. What is new here is not the readings but that each one is *priced*. Every row sets out what its reading would require a scribe to have done, and gives the rate at which the corpora show scribes doing it, so that a reader who doubts a reading can see which step to attack and what it would cost to defend. The same treatment is applied to proposals that are not ours, and it is applied to this author without exemption: several readings here were withdrawn while the file was being written because a measurement ruled them out, and §S1.14 records which. A reading that survives that is still not a finding. But it is a different object from a resemblance, and that difference is what the file is for.

Section 9 of the paper sets out what a proposed reading of a Xiongnu name has to survive before it can be called a finding: the sense has to be attested in a source independent of the comparison, the morphology has to be licensed by a suffix with the right distribution, and any objection that can be tested has to be tested. The file covers all forty items of the record, and they are not all of one kind. **One** reading is argued in the paper itself and meets its standard: 若鞮 as *İnakt* (§9). **One** carries a comparison taken from the literature rather than made by us, which the paper then tests: 徑路 as *Kıngrak* (§9.1). **One** is a compound pointing at its constituent rows. **Two** now carry no reading at all. At 羯 the proposal was withdrawn and the reading in print left standing; at 呼都而尸道皋若鞮 it was withdrawn on three measurements and nothing put in its place, and the word it had claimed stays at 烏累若鞮, where the transcription supports it. The remaining **thirty-five** are the author's own proposals and none of them meets the standard, 冒頓 among them: §9.2 of the paper measures the three values its second character can take and does not choose, and the proposal made here is set out in S1.4 with its objections.

They are here since we produced them over the course of the work. A reader who has seen **稽粥** written out as *kei çuk* will hear Turkish *küçük* whether or not this supplement exists. What can be done is to write them down, date them, label them, and put next to each one the reason it occurred to us, so that the reasoning is inspectable rather than implied.

Nothing in the paper depends on this supplement. The measurements in §§5–8 (the channel accuracies, the period-gap decomposition, the initial-consonant probe, the coda-spelling count) were made without reference to these readings and do not change if every one of them is rejected.

A dagger, †, on a syllable in the tables below marks a **polyphonic character**, one for which Schuessler's table gives more than one Later Han reading. The reading printed is the first tabulated one, as in Appendix A of the paper. Table S7 lists every such character with the readings not used, and where an alternative would change a proposed reading the entry for that item says so.

S1.2 Why they are not in the paper

Section 6.3 measures exactly the operation that produced this supplement, taking a reconstructed reading and looking it up in a Turkic dictionary, against a null model built by resampling Later Han syllables into pseudo-names. Across seven lexicons, from modern Turkish to the *Codex Cumanicus*, between none and three names in thirty-six meet significance of $p \le 0.05$. At a one-in-twenty threshold, thirty-six coin flips give the same.

Doing the matching by hand does not improve on the algorithm. It makes it worse. A person cannot help preferring the reading that means something, and that preference is precisely the selection effect the null exists to quantify. The algorithm at least fails honestly. A paper that measured the unreliability of this operation and then presented its own outputs as discoveries would be incoherent.

So we keep the two things apart. The paper reports what can be measured. This file records what we imagined while measuring it. Because it is supplementary, it does not carry the paper's evidential standard and should not be read as though it did.

S1.3 How the readings were produced

Every row starts from the same two published inputs, neither of them model output. The Chinese form is from the *Shiji*, *Hanshu* and *Hou Hanshu*. The Later Han reading is Schuessler's. The third column simply respells that reading in Turkish orthography, so that a reader who does not read phonetic notation can see what is being claimed. It adds no information.

The fourth column is ours. We arrive at it by hearing the third column as a Turkic word or compound, usually with a vowel changed, sometimes treating a consonant as lost, and then looking for a historical parallel that would make such a name plausible for a steppe ruler. The fifth column gives the sense that reading would carry. The note beneath each row gives the parallel. Each entry below is set out in two parts: *Phonetics*, which walks the Later Han reading to the proposed name character by character and says which of the steps are ordinary correspondences and which are not; and *Reason*, which gives the word, its attestation where it has one, and the naming parallel that suggested it.

The standard we used for inclusion here is deliberately lower than the paper's, and it is worth stating. A reading is offered when an attested word fits the transcription and nothing in the transcription stands against it. It is not offered when the spelling rules it out, when a character would have to carry a sound the corpora never show it carrying. Under-determination is not a reason to withhold a reading, because almost every short name in the record is under-determined; a demonstrable phonological conflict is a reason.

We had three reasoning habits. We list them because they are the main weakness. The first is the epithet argument: later Turkic rulers are commonly named by a common noun or a natural force: Bayezid I *Yıldırım* "the thunderbolt", Alp Arslan "the lion", Mehmed IV *Avcı* "the hunter", so a reading that yields such a noun is treated as more likely. The second is the career argument: where the Chinese record says what a ruler did, a reading that names the deed is preferred. The third is the compound argument: a long transcription is split into two or three Turkic elements. Each of these makes a reading easier to reach, and none of them makes it more likely to be right. They multiply the search space, which is the opposite of evidence.

S1.4 The reading argued in the paper, and this author's proposal for 冒頓

Two items in this supplement are set out at length. The first is argued in the paper rather than merely recorded, because it is supported differently from everything else in the file: the recurring element **若鞮** `ńak te`, which closes the formal names of six later chanyu, read *İnakt*, "the faithful ones" (§9):

- the *Hou Hanshu* glosses **若鞮** as **孝** "filial", the meaning is recorded in the source, not inferred from a resemblance;
- *inak* is attested as *fidelis* "faithful" in the *Codex Cumanicus* (Kuun 1880, p. 182), a medieval source with no connection to the Chinese record;
- the morphology is licensed: `-t` is an Old Turkic plural used with titles (*tegin* → *tegit*, *tarqan* → *tarqat*), and the competing `-tI` reading we ruled out by counting: of 31 Cuman headwords in `-tI`, almost all are past-tense verbs;
- we tested the obvious objection to the spelling and it failed: if the source ended in `-t`, why did the scribe use the open character **鞮**? Because 30% of stop-final source syllables in the Sanskrit corpus were written with an open character.

One objection stands and is stated in the paper: if the source word was *ınakt*, the initial *ı-* is not written, and **若鞮** has no room for it.

The second is not a reading the paper argues. 冒頓 `mək tuən`^{c} is the founder's name, and §9.2 prices the three values its second character can take without choosing among them. **This author's proposal is *Bögü Tuğ***, "sage standard", and it is set out here rather than there, with the objections to it, because that is where proposals in this work belong.

- **The frame.** Both codas are written, which makes it one of the tightest frames in the record, but neither character has a single value. 冒 has two Later Han readings, `mək` and `mou`^{c}; 頓 has `tuən`^{c} and `tuət`, and a third value from the commentarial note 頓音毒, which Pulleyblank reports as Early Middle Chinese `dawk`. That third value is a substitution recorded by a commentator, not a row of the table, and the paper labels it as such.
- **The first element.** 冒 and 默 share the reading `mək`, and 默啜 is read *Bögü-Çor*, the personal name of Qapaghan Qaghan; 牟 `mu` writes the same word in 牟羽 for the Uyghur Bögü Qaghan. So both rows of 冒 have a precedent for writing *bögü*. Table 9 of the paper prices them: a rounded source vowel is written with a rounded Chinese vowel in 59% of cases and an open one in 4%, so the second row is where *bögü* is ordinary rather than merely permitted.
- **The second element.** *tuğ* is the horse-tail or yak-tail standard carried on a pole, one of the oldest continuously used objects of steppe rulership. It is attested in Old Turkic: *tuğ taşıkar* "when the standard went out" in the Moyun Çor inscription, cited by Doerfer at ETY I 170, and Kāşgarî glosses it as the drum and banner borne before the king, with *toquz tuğluğ xan* "a khan of nine standards". It ends in a velar, so it needs 頓's second row (2 of 21 opportunities) or the substitution (10 of 21), and it is excluded on the first row (0 of 21).
- **The count that follows.** Two of the three values of 頓 admit a velar-final second element, one admits the liquid-final *Bayatur* the field has read for a century, and one admits a nasal-final one. That is the whole of the arithmetic, and it is why this author prefers a velar-final reading.
- **An observation about the substitution.** In the Later Han table 毒 and 纛 are exact homophones, `douk`, and 纛 is the Chinese word for the yak-tail standard itself. So the commentator who wrote 頓音毒 put the value of 頓 at precisely the Later Han reading of the word for the standard. This may be coincidence, and one homophone pair proves nothing, but it is recorded here rather than left out.
- **The name shape.** *bögü* is a regnal epithet across the record: *Teŋri Bögü Teŋriken* in the Old Turkic inscriptions, Bögü Qaghan of the Uyghurs (牟羽), Bögü-Çor of the Türks (默啜). Kāşgarî glosses *bügü* as "bilgin, akıllı, hakim"; Clauson says the word "seems to connote both wisdom and mysterious spiritual power", and it was borrowed early into Mongolian as *böge*, the ordinary word for a shaman. Two nouns in apposition is the commonest Turkic name pattern.

The objections, all of which stand.

- **The loan direction.** Doerfer's *TMEN* gives Turkic *tuğ* as a loan from Chinese (entry 969). If that is right, the word cannot be an element of a name of the early second century BC, and the reading fails. Doerfer's own addendum in volume IV weakens it: it quotes Menges (1968: 169, as cited there) that the Chinese word "might well be an ancient loanword in Chinese", and H. Franke, in a letter of 23 March 1966, that the Chinese word is first attested in the *Zhouli* in a purely Chinese context and is "kaum jünger als 2. Jh. v. Chr.", which is the founder's own century. The direction is therefore disputed by the same authority usually cited for it, and this author does not settle it.
- **The row it needs.** The reading is ordinary on the substitution, marginal on 頓's second row, and excluded on the first, which is the row Appendix A of the paper prints. It also reads better on the second row of 冒 than on the first, which Appendix A likewise prints.
- **The compound is not attested.** Only its two halves are, as with every compound reading in this supplement.

- **The frame does not fix the vowel** of the second syllable, so *tuğ* is what the velar coda and the Turkic lexicon together suggest, not what the transcription requires.

The rival reading is not free of these either, and it is fair to say so. Clauson's entry on *bayatur* records that the word "occurs only once in the early period and then still as a P.N.", that the sense "picked warrior" most probably developed in Mongolian rather than Turkic, and that Turkic had other words such as *alpayut* in that meaning; Doerfer's entry 817 adds that the form *bayatur* is not findable in the Old Turkic dictionaries and texts available to him, the attested Old Turkic form being *bātur*, and that the reconstruction `*bayatur` is secured by Old Hungarian *bahatur* rather than by a Turkic attestation. A form without a medial velar leaves 冒's written coda with nothing to write.

Both appear in the tables below in their proper places, so that this supplement can be read as one list with the standard visible in it. The second appears there as an author's proposal, which is what it is.

S1.5 Why the decoder's output is not cited below

The channel of §4 has a decoding step that will return a ranked list of readings for any string in the record. None of those rankings is quoted in this supplement, and the reason is worth stating, because the numbers look more informative than they are.

A character can be *reading-anchored*, meaning its emission distribution was learned for that character specifically, or it can fall back to a table keyed on the Later Han **initial consonant class alone**, discarding the vowel and the coda. Of the 39 items in the record, only five are reading-anchored throughout; 34 use backoff on at least one character and nine use it on every character. The effect is that whole groups of characters share one distribution:

LATER HAN INITIAL CLASS	CHARACTERS IN THE XIONGNU RECORD SHARING ONE EMISSION DISTRIBUTION	WHAT THE BACKOFF TABLE RETURNS FOR EVERY ONE OF THEM
velar	17: 稽 渠 胡 湖 壺 戶 侯 姑 孤 骨 谷 韓 諧 皋 權 朐 徑	ka 0.34, ga 0.15, ku 0.08
affricate	6: 且 粥 珠 臣 詹 郅	ca 0.18, ja 0.10, cā 0.08
dental stop	9: 鞮 題 當 屠 撐 株 稚 脫 頓	ta 0.23, da 0.11, ti 0.10
liquid	6: 廬 攣 祿 累 蠡 閭	ra 0.37, la 0.17, rī 0.08

TABLE S1 Because 稽 k^{h}ei^{B} and 渠 gɨɑ draw on the same row, and 粥 tśuk and 且 tsɑ^{B} on another, an affricate-plus-velar string and a velar-plus-affricate string return mirror images of one table entry rather than anything specific to those characters. Two further limits apply: the scores are normalised over a retained beam, so they are relative rankings within it and not posteriors; and the syllable inventory is Sanskrit-derived, so q and the front rounded vowels ö and ü cannot be produced at all, so a reading containing ü is therefore not rejected by the decoder, it is unavailable to it. §6.2 of the paper makes the general point; this table is its mechanism.

In the entries below, where a spelling question arises, we cite the paper's initial-correspondence counts (§8.1): direct counts of which Chinese initial was used for which source sound across the three corpora. Those are observations rather than model output, and they are what the word "permitted" means wherever it appears in this supplement.

S1.6 Chanyu, in order of reign

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
頭曼 Touman	do mɑn^{c†}	do man	**Duman**	dust cloud
PHONETICS 頭 *do* → *du*: the initial is direct; o and u are not separable at the ~50% vowel recovery §7 reports. 曼 *mɑn* → *man*: exact, and the coda -n is written, because -n is one of the six codas the Chinese syllable can carry. One vowel change in two syllables. The rival reading *tümen* needs ü, which the Later Han script cannot write at all. **REASON** *duman* / *tuman* / *touman* “smoke, fog, mist” is one word across Kāşgarî, the Codex Cumanicus and modern Turkish. The standard identification is *tümen* “ten thousand”, which is absent from the Cuman vocabulary entirely.				
冒頓 Modu / Modun	mək† tuən^{c†}	mık tuın	**Bögü Tuğ**	sage standard

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING

PHONETICS The frame is **b/m – k/g – t/d – nasal or stop**, and **both codas are written**, which makes it one of the tightest frames in the record and also one of the least determinate, because neither character has a single value. 冒 has two Later Han readings, *mək* and *mou*^{c}; 頓 has *tuən*^{c} and *tuət*, and a third value from the commentarial note 頓音毒, reported by Pulleyblank as Early Middle Chinese *dawk*, which is a substitution recorded by a commentator rather than a row of the table. 冒 *mək* → *bögü*: the m- stands for a source b-, 3 of 22 in the Later Han corpus, 13.6%; the written -k stands for the source g; and the final vowel is not written, one character covering two source syllables, a device found in 47 of the 1,017 verified pairs, as 竭 covers *-gata* in *tathāgata*. There is a character-level precedent as well as a rate: 冒 and 默 share the reading *mək* and 默啜 is read *Bögü-Çor*, while 牟 *mu* writes the same word in 牟羽 for the Uyghur Bögü Qaghan. 頓 → *tuğ*: a velar-final second element, which the corpus writes with the -t coda of 頓's second row in 2 of 21 opportunities and with the -k of the substitution in 10 of 21, and never with the -n of the first row, 0 of 21.

REASON **Bögü Tuğ, "sage standard".** *tuğ* is the horse-tail or yak-tail standard carried on a pole, attested in Old Turkic (*tuğ taşıkar* "when the standard went out", Moyun Çor, cited by Doerfer at ETY I 170) and glossed by Kāşgarî as the drum and banner borne before the king, with *toquz tuğluğ xan* "a khan of nine standards". *bögü* is a regnal epithet across the record: *Teŋri Bögü Teŋriken*, Bögü Qaghan of the Uyghurs, Bögü-Çor of the Türks; Kāşgarî glosses *bügü* as "bilgin, akıllı, hakim" and Clauson says it "seems to connote both wisdom and mysterious spiritual power". Both halves are attested independently of any Chinese source and two nouns in apposition is the commonest Turkic name pattern. **Why a velar-final second element.** Of the three values 頓 can take, two admit a velar-final element, one admits the liquid-final *Bayatur* the field has read for a century, and one admits a nasal-final one. §9.2 of the paper sets out that comparison and does not choose; the preference here is this author's. **What is not settled.** The loan direction of *tuğ* is disputed: Doerfer's *TMEN* 969 gives it as a loan from Chinese, which would put it out of reach of a name of the early second century BC, while his own addendum in volume IV quotes Menges (1968: 169, as cited there) that the Chinese word "might well be an ancient loanword in Chinese" and Franke (letter of 23 March 1966) that the Chinese word is first attested in the *Zhouli* and is "kaum jünger als 2. Jh. v. Chr.". A ninth-century check adds nothing either way, and is recorded so that it is not made twice: Tekin's glossary to the *Irk Bitig* has the verb *tug-* "to rise (of sun)" but not the noun, the text's word for a flag is *uçrug*, and *bögü* is not in it. That text is sixty-five omens and a few hundred words, so absence from it is not evidence of anything. The reading is also ordinary on the substitution, marginal on 頓's second row and excluded on the first, which is the row printed here; the compound is not itself attested, only its halves; and the frame does not fix the vowel of the second syllable. The rival reading carries its own difficulties, recorded in S1.4. Full discussion in S1.4.

稽粥 Jizhou / Jiyu	k^{h}ei^{B}† tśuk	kei çuk	**Kiçük**	the younger

PHONETICS 稽 *k*^{h}*ei* → *ki*: an aspirated velar is the ordinary vehicle for a source k, and the diphthong *ei* for *i* is routine. 粥 *tśuk* → *çük*: tś- is the default character class for a source affricate (37% in the Sanskrit corpus), and the coda -k is actually written, the strongest part of the reading, since a stop coda goes unwritten 30% of the time. Only ü is unavailable and surfaces as u.

REASON *kiçig / kiçik / kiçük* "small, younger" is attested three times independently: Kāşgarî has *kiçik* and *kiçük* with the derivative *kiçiglik*, the *Kutadgu Bilig* has *kiçig oğlan erken* "when he was a small boy" (1823), and the Codex has *Kečak* glossed *paucus*. 稽粥 is the personal name of Laoshang chanyu, Modu's son and successor, so "the younger" fits the succession the Chinese sources record.

軍臣 Junchen	kun dźin†	kun cin	**Köngen**	the one who is always upright

PHONETICS 軍 *kun* → *kön*: three steps, all direct. A source velar written with a Chinese *k-* is 190 of 220, 86.4%; a source rounded vowel written rounded is 190 of 289, 65.7%; the *-n* is written, 77 of 93, 82.8%. 臣 → *-gen* takes the second of the character's two Later Han readings, the velar *gin*, rather than the palatal *dźın* that Appendix A prints: a source *g* written with a Chinese *g-* is 52 of 90, 57.8%, and a source front vowel written with a front vowel is 327 of 380, 86.1%. **The palatal reading is not available to this name.** A source voiceless affricate written with the voiced *dź-* is 0 of 48 in the Sanskrit corpus, so *çin*, *çen* and *çan* are excluded outright; the Codex forms that appear to offer an affricate are an artefact of medieval Latin spelling, in which *j* writes Turkic *y-* and *c* before a back vowel writes *k-*. Step 47 prices every combination the two characters permit, and *köngen* is the cheapest at 19.4%, about 1 in 5, against 1 in 9 for the best branch of 冒頓 and the 1 in 37 that retires *Bayatur* in §8.1 of the paper. Two syllables, two characters, nothing inserted and nothing discarded.

REASON **Köngen, "the one who is always upright".** The stem is the verb *kön-*, which Kāşgarî glosses *düzelmek, doğrulmak; yola gelmek*, to straighten, to become upright, to come to the right way, and which carries a family around it: *könitmek*, *köndgermek* and *köndgürmek*, all *doğrultmak*. The suffix is *-gAn*, the deverbal participle with a habitual force. **The suffix is counted here, not assumed.** Across the 9,379 headwords of the Kāşgarî index, the Codex Cumanicus and Tekin's glossary to the *Irk Bitig*, *-gAn* has 211 headwords and 109 of them are glossed with *daima*, *her zaman* or *çok*: *arıtgan* "her zaman temizleyen", *egilgen* "daima eğilen", *etilgen* "her zaman düzelen", which is the near sibling of the form proposed here. The competing suffixes *-gUn* and *-gIn* have 72 headwords between them and not one habitual gloss: they are concrete nouns, birds and plants and body parts, a knot, a bridle. In the eleventh-century record *-gAn* is the productive participle and *-gUn* is not. The modern *durgun*, *coşkun* and *ötgün* belong to a later productivity of *-gUn* and are not evidence for this period. **Vowel harmony decides a tie the transcription cannot.** *Köngin* costs exactly what *köngen* costs, 1 in 5, because 臣 carries a front vowel either way. But *-gAn* has two shapes while *-gIn* and *-gUn* have four, so a front rounded stem such as *kön-* must take *-gün* and cannot take *-gin*. The Turkic evidence removes a candidate that the Chinese evidence cannot. **What is not settled.** *Köngen* is not itself an attested headword. It is a formation by a productive rule from an attested stem, which is a stronger position than a form requiring a segment that must afterwards be lost, but it is not an attestation. The lexicons are eleventh century and the name is second century BCE, so that *-gAn* was productive for Kāşgarî does not establish it 1250 years earlier. Whether *-gAn* formations were used as personal names or titles in Old Turkic is not tested here. The *g* to *g* rate is measured over all positions, and whether a *g* after a nasal behaves the same has not been measured. And the first element is undetermined: *kön*, *kün*, *kon* and *kun* all cost the same 47.0%, so choosing *kön-* is a choice made on meaning, which is the first of the three reasoning habits this supplement lists against itself.

詹師廬 Zhanshilu	tśam ṣi liɑ	çam ṣi lia	**Çamşılı**	dazzling, radiant

PHONETICS 詹 *tśam* → *çam*: tś- is the default affricate spelling and the -m coda is written. 師 *ṣi* → *şı*: the retroflex sibilant is the standard vehicle for ş. 廬 *liɑ* → *lı*: note that Chinese l- also carries source r-: in the paper's correspondence counts an r- was written l- in 12 of 14 attestations, so this syllable is genuinely ambiguous between -lı and -rı.

REASON A stem of shining or dazzling. The solar reading this once echoed at 軍臣 has been withdrawn, so it stands alone.

<table>
<tr><th>FORM, AS WRITTEN IN THE HISTORIES</th><th>LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)</th><th>THE SAME READING IN TURKISH ORTHOGRAPHY</th><th>READING WE PROPOSE</th><th>SENSE WE PROPOSE FOR THAT READING</th></tr>
<tr><td>呴犁湖
Xulihu / Goulihu</td><td>huo† lei† gɑ</td><td>huo lei ga</td><td>Kalık</td><td>sky, heaven</td></tr>
<tr><td colspan="5">PHONETICS 呴 huo → ka, 犁 lei → lı, 湖 gɑ → k: three onsets, each taken from the majority correspondence, with nothing inserted and no coda discarded. The velar class covers h-, k- and g- alike, so the first and third syllables cost nothing.
REASON Kâşgarî glosses kalık “hava, gök, sema” (air, sky, heaven). The word belongs to the same semantic field as 撐犁, the one Xiongnu term the Chinese sources gloss (天 “heaven”), so a ruler bearing it would not be isolated in the record. A second exact fit, kılık “disposition, manner”, is available if the sky reading is thought too convenient.</td></tr>
<tr><td>伊稚斜
Yizhixie</td><td>ʔi ḍi^{C} ja</td><td>’i di ya</td><td>İdiya</td><td>master</td></tr>
<tr><td colspan="5">PHONETICS 伊 ʔi → i: a glottal onset is the standard way to write a vowel-initial syllable. 稚 ḍi → di: retroflex ḍ for d is routine. 斜 ja → ya: this character class emits ya more strongly than any other single-syllable correspondence in the record. All three steps are default.
REASON idi “lord, master”, attested in Old Turkic. Compare Murad I's epithet Hüdavendigâr, “the sovereign”, a title of the same shape and force.</td></tr>
<tr><td>烏維
Wuwei</td><td>ʔɑ wi</td><td>’a vi</td><td>Abı</td><td>one who hunts or stalks</td></tr>
<tr><td colspan="5">PHONETICS 烏 ʔɑ → a: vowel-initial, direct. 維 wi → bı is the weak step: the Chinese character carries a w-, and while Turkic b- and w- alternate, the transcription reads more naturally as avı than abı. This is one of only five items in the record decoded entirely from character-specific evidence, so the mismatch cannot be attributed to a gap in the data.
REASON av “hunt”, avcı “hunter”. Avcı names an occupation; abı is proposed as naming a trait. Avcı is also the epithet of Mehmed IV.</td></tr>
<tr><td>且鞮侯
Qiedihou</td><td>tsɑ^{B†} te† go</td><td>tsa te go</td><td>Çatıgka</td><td>one ready for the clash</td></tr>
<tr><td colspan="5">PHONETICS 且 tsɑ → ça: ts- for a source affricate is a minority spelling but a real one, 15% in the Sanskrit corpus and 9% in Ligeti. 鞮 te → tı: direct. 侯 go → gka asks a single Chinese syllable to carry a consonant cluster it structurally cannot, so the reading adds material the transcription does not have.
REASON çat- “to strike together, to clash”. Compare Selçuk Bey, whose name has been read as “perpetually ready for the fight”.</td></tr>
<tr><td>狐鹿姑
Hulugu</td><td>ɣuɑ lok kɑ</td><td>ğua lok ka</td><td>Kurluga</td><td>the binder, the encloser</td></tr>
<tr><td colspan="5">PHONETICS 狐 ɣuɑ → kur: a velar fricative for a Turkic back q, which §8.5 of the paper measures at 43 of 66, 65.2%, against 20 of 66 for an aspirated character; kurluga has back vowels throughout. The rounded vowel is written rounded, 190 of 289, 65.7%. The -r is not written, and that is the ordinary treatment rather than a difficulty: Table 7 gives a source liquid coda spelled with an open character 21 times in 37, 56.8%, the majority option, and 狐 is open. 鹿 lok → lu: a source l written with a Chinese l- is 85 of 88, 96.6%, the vowel rounded again. Its written -k is not discarded. The character after it is 姑 kɑ, a velar, and a medial written coda writes the onset of the following syllable in 165 of 446 cases, 37.0%, which is the largest single thing such a coda does: 僧 for saṃ-ghā-, 三 for saṃ-, 怛 for ta-thā-. 姑 kɑ → -ga, with the vowel open, 1045 of 1509, 69.3%. The whole name costs 1.90%, about 1 in 52, against the 1 in 37 that retires Bayatur in §8.1. The weak step is the last one, not the first two. A source g written with a Chinese k- is 12 of 90, 17.8%. Were the word kurluka the same character would be writing a source k at 86.4% and the name would cost 1 in 10, so one consonant moves the price fivefold. Step 51 sets this out.
REASON Kurluga, from the sense of binding and enclosing. The root is attested and so is the sense: Kâşgarî has kur “kuşak, kemer”, a belt or girdle, and the verb kurlamak, “kuşak yapmak ve bağlamak”, to make a belt and to bind, together with kurluk and kurlanmak. 狐鹿姑 captured the Han general Li Guangli and held the confederation together through a hard decade, so an epithet of binding fits what the record says he did. Compare the Göktürk ruler Kapgan Kagan, kapgan “the encloser”. Two objections in an earlier draft of this entry were wrong and are withdrawn. It described the -r as a segment Later Han cannot write and the -k as a discarded coda, and said the two pulled in opposite directions. Neither holds: leaving a liquid unwritten before an open character is the majority spelling at 57%, and a written coda reaching forward to the next onset is the commonest use of a written coda at 37%. Both were judged by ear before they were counted. What is not settled, and it is the shape of the word rather than any of its sounds. Kurluga is not an attested headword, and neither is kurluka; the frame returns no word at all. More than that, the shape is not one Turkic builds: the suffix -lXk / -lXg is word-final, 380 headwords end in it, while only one headword in 9,343 ends in -luka and none in -luga. The well-formed Turkic word in this frame is kurluk or kurlug “girded”, which 狐鹿 write between them with 鹿's own -k as the final consonant, and which leaves 姑 unaccounted. That is the same structural question as at 虛閭權渠: whether a longer Xiongnu name is one word or a name followed by a title element. It is not answered here, and this reading assumes the first.</td></tr>
<tr><td>壺衍鞮
Huyandi</td><td>gɑ jan^{B†} te†</td><td>ga yan te</td><td>Kayantı</td><td>avalanche</td></tr>
<tr><td colspan="5">PHONETICS 壺 gɑ → ka: given a Later Han voiced g-, the source sound is more often k than g, so the devoicing is normal rather than a liberty. 衍 jan → yan: direct, coda written. 鞮 te → tı: direct. All three steps are default or better, phonetically one of the cleaner readings here.
REASON kay- “to slide, to slip away”. Parallel in sense to Bayezid I's Yıldırım: a natural force taken as a regnal epithet.</td></tr>
<tr><td>虛閭權渠
Xulüquanqu</td><td>k^{h}ɨɑ liɑ gyan gɨɑ</td><td>kıa lia gyan gıa</td><td>Kalkan</td><td>shield</td></tr>
</table>

<table>
<tr><th>FORM, AS WRITTEN IN THE HISTORIES</th><th>LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)</th><th>THE SAME READING IN TURKISH ORTHOGRAPHY</th><th>READING WE PROPOSE</th><th>SENSE WE PROPOSE FOR THAT READING</th></tr>
<tr><td colspan="5">PHONETICS 虛 kʰɨɑ → ka, 閭 liɑ → l, 權 gyan → kan: the first three characters match exactly, including the written -n coda, at zero minority steps. Only 渠 gɨɑ is left unaccounted for, and it may carry a separate title element.
REASON kalkan “shield” is in Kâşgarî and in the Codex Cumanicus as Kalkani, “scutum ejus”. A weapon or armour name is an ordinary steppe formation, and it needs no compound to be read.</td></tr>
<tr><td>握衍朐鞮
Woyanqudi</td><td>ʔɔk jan^{B†} guo te†</td><td>’ok yan guo te</td><td>Okyan Kutuğtı</td><td>blessed sovereign of the arrow flank</td></tr>
<tr><td colspan="5">PHONETICS 握 ʔɔk → ok is exact, with the -k coda written, and 衍 jan → yan and 鞮 te → tı are direct. The weak point is 朐 guo → kutuğ, which expands one Chinese syllable into two Turkic ones; nothing in the corpora licenses that expansion.
REASON ok “arrow” doubles as a division of the confederation in later Turkic usage (the On Ok, “ten arrows”, of the Western Turks) which is what makes a territorial reading available. kut is “heavenly favour, fortune”.</td></tr>
<tr><td>呼韓邪
Huhanye</td><td>hɑ† gɑn ja†</td><td>ha gan ya</td><td>Kaganya</td><td>kagan of the flank</td></tr>
<tr><td colspan="5">PHONETICS 呼 hɑ → ka: a source back q written with an h- character, which §8.5 of the paper measures at 43 of 66, 65.2%, against 20 of 66 for an aspirated character. It is the majority spelling, not a substitution needing indulgence. The 27% figure earlier drafts quoted here was the Mongolian rate; the Turkic one is the relevant measurement and it is higher. 韓 gɑn → gan: exact, coda written. 邪 ja → ya: a source y- takes this character class 57 times in 98, 58.2%, the largest single destination for a source y- though not a majority of the corpus. The written sequence ha gan ya becomes ka-gan-ya on one change.
REASON 呼韓邪 commanded the left flank before becoming ruler, so the Chinese record itself supplies the office that the reading names.</td></tr>
<tr><td>郅支
Zhizhi</td><td>tśit tśe†</td><td>çit çe</td><td>Çitçi</td><td>border-maker, one who carves the boundary</td></tr>
<tr><td colspan="5">PHONETICS 郅 tśit → çit: tś- is the default affricate vehicle and the -t coda is written. 支 tśe → çi: default. Both syllables use the majority correspondence and neither requires an inserted or deleted segment.
REASON çit / çiz-, of marking or carving a boundary. 郅支 broke away westward and set up a rival seat, so a name of division fits the career, though it is equally possible that the career suggested the name.</td></tr>
</table>

TABLE S2 The fifteen rulers whose names are not built on the recurring element 若鞮. Reign order follows the *Shiji* and *Hanshu* as reproduced in Appendix A of the paper.

S1.7 The six 若鞮 compounds

<table>
<tr><th>FORM, AS WRITTEN IN THE HISTORIES</th><th>LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)</th><th>THE SAME READING IN TURKISH ORTHOGRAPHY</th><th>READING WE PROPOSE</th><th>SENSE WE PROPOSE FOR THAT READING</th></tr>
<tr><td>若鞮
Ruodi</td><td>ńɑk te†</td><td>nyak te</td><td>İnakt</td><td>the faithful ones</td></tr>
<tr><td colspan="5">PHONETICS 若 ńɑk → ınak: a palatal nasal onset with the -k coda written. 鞮 te → -t: the final stop is not written, and Table 7 of the paper licenses that at 30%. One objection stands and the paper states it: if the source was ınakt, the opening ı- is not written and 若鞮 has no room for it.
REASON One of the two readings argued in the paper itself, at §9, and it is listed here so this supplement can be read as a single list with the standard visible in it. Inak is glossed fidelis “faithful” in the Codex Cumanicus; the Hou Hanshu glosses 若鞮 as 孝 “filial”; the plural in -t is an Old Turkic formation used with titles (tegin → tegit); and Clauson records Chagatay inak as “a royal representative or senior minister”, so a court title on this root is what the word ordinarily does.</td></tr>
<tr><td>復株累若鞮
Fuzhulei Ruodi</td><td>buk† ṭio lui† ńɑk te†</td><td>buk tio lui nyak te</td><td>Büktel İnakt</td><td>Büktel, of the İnakt</td></tr>
<tr><td colspan="5">PHONETICS 復株累 buk ṭio lui → büktel: the b-, the -k coda, the dental and the liquid all fall in place across the three characters, at zero minority steps, with only the front rounded vowel unwritable. One attested word covers the whole prefix rather than an invented two-element compound. 若鞮 → İnakt is treated in §S1.4.
REASON Kâşgarî gives büktel as “orta boylu; yassı arkalı, oturamaklı”: of middling height and flat-backed. A name from a bodily description is a well-attested Turkic type; compare the Hungarian royal name Şişman.</td></tr>
<tr><td>搜諧若鞮
Souxie Ruodi</td><td>so† gɛi ńɑk te†</td><td>so gei nyak te</td><td>Sarıg İnakt</td><td>the yellow, the fair, of the İnakt</td></tr>
</table>

<table>
<tr><th>FORM, AS WRITTEN IN THE HISTORIES</th><th>LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)</th><th>THE SAME READING IN TURKISH ORTHOGRAPHY</th><th>READING WE PROPOSE</th><th>SENSE WE PROPOSE FOR THAT READING</th></tr>
<tr><td colspan="5">PHONETICS 搜 so → sar: a sibilant onset, which the correspondence counts make obligatory: a source k, q or g was written with a Chinese sibilant in 0 of 2,124 attestations, while Chinese sibilant characters transcribe s, š, č or ś in 98% of 837 cases. The final -r is not written and could not be: Later Han carries no -r in any coda position. 諧 gɛi → ıg: a velar onset, direct. Nothing is inserted, and no character is read as a sound it cannot carry. 若鞮 → İnakt is treated in §S1.4.
REASON sarıg “sarı, sarı renk” (yellow) is in Kāşgarî, and survives as modern Turkish Sarı, still current both as a name and as an epithet. Color epithets are a documented Turkic naming type: the same dictionary has çakır “grey-eyed”, and Çal “dappled, grey” is proposed for 烏珠留若鞮 on the same footing. Ninety-one attested headwords fit this two-syllable frame, so it does not determine the reading; sarıg is offered as the most plausible of them, with nothing in the transcription standing against it.</td></tr>
<tr><td>車牙若鞮
Cheya Ruodi</td><td>tśʰa† ŋa ńɑk te†</td><td>ça nga nyak te</td><td>Tunga İnakt</td><td>the tiger, or the hero, of the İnakt</td></tr>
<tr><td colspan="5">PHONETICS 車 tśʰa → to: the first of the character’s two tabulated readings, no polyphony argument needed. A source t written with the aspirated tśʰ- is 67 of 198, 33.8%; the character is open, so any source coda would be unwritten; the vowel is free, a source o written with an open vowel being 86 of 289, 29.8%. 牙 ŋa → -ŋa: a source syllable beginning with a velar nasal takes a Chinese ŋ- character 5 times in 7, and the one intervocalic case is 丁彥多 for tiṅanta, where 彥 carries the -ṅan-; a source a written with an open vowel is 1045 of 1509, 69.3%. The whole name costs 4.98%, about 1 in 20, against the 1 in 37 that retires Bayatur in §8.1 of the paper. Step 48 prices every form the two characters permit. The reading this replaces is excluded twice over. Ok required 牙 to write a vowel-initial syllable, which is 0 of 140 in the Sanskrit corpus, 137 of the 140 taking a glottal; and çöge required a source affricate written with the aspirated tśʰ-, which is 1 of 54, where the vehicle for a source affricate is the plain tś- at 32 of 48. It also added a syllable the character does not carry. A cheaper form exists and is rejected on sense, not on phonology. On the second reading of 車 the frame admits kaña, from the Codex canga, at about 1 in 3; it is glossed tabulatum, planking, which is not a ruler’s name, and it needs the second reading where toña needs only the first. The transcription underdetermines here as it does everywhere, and the choice is made on meaning, which §6.3 of the paper measures as the weakest step in the whole procedure. On the spelling. The -ng- of Turkish Tunga renders a single velar nasal, not a nasal followed by a stop. That is why these two characters write the word and 當戶 cannot: a written -ŋ coda followed by a velar onset is the corpus device for a source ŋg or ŋk cluster, 13 cases of 13, and this word has no velar stop in it.
REASON Tunga, an epithet, in the title slot the other 若鞮 names use. Clauson’s headword is toŋa:, with back vowels, and he records that “whether -o- or -u- in the first syllable is uncertain”, so Tunga and Tonga are equally available; this file prints the first, and the transcription cannot separate them either, since 車 carries an open vowel and a source o and a source u fall alike at 29.8%. Two senses, resting on different evidence, and both are recorded here. Kāşgarî glosses the word as a tiger-like beast (Dīwān III 368), and glosses the title Toŋa Alp Er, given to Afrasiyab the king of the Turks, as “the heroic man as strong as a tiger”. Outside personal names the eighth- and ninth-century Uyghur texts use it for “hero, outstanding warrior”: the Manichaean éligler xanlar toŋalar, “kings, khans, great warriors” (M III 41), the Buddhist Arcunı toŋa, “the hero Arjuna” (U II 24), and toŋalar begi. Clauson records that the animal sense “is not confirmed by any other authority”, and that by Kāşgarî’s day the word’s meaning had become unknown and it survived chiefly as a title. Neither sense is suppressed here, and the choice between them does not change the reading. The naming pattern is attested, and it is the one this name needs. Kāşgarî says the word is used as a laqab, an honorific epithet, and gives toŋa xan and toŋa tegin: the epithet first, the title after it, head last, which is the shape of Alp Arslan and Bilge Kağan. Tunga İnakt is that shape, with 若鞮 as the title element the paper argues in §9, and it matches the form the other five 若鞮 rows already take, such as Çal İnakt. What is not settled. The vowel of the first syllable, which neither Clauson nor the transcription fixes. Whether the epithet is the animal or the quality, which turns on how much weight Kāşgarî’s gloss can bear given that he reports the word as already opaque. The Dīwān’s own text is not held for this project, only the Türk Dil Kurumu index, which carries headwords without verses, so the order Alp Er Tonga familiar from the elegy could not be checked here: Clauson gives only Toŋa Alp Er, from Dīwān III 368 and Kutadgu Bilig 277. And this word stood at 當戶 in earlier drafts; it is moved here because 當戶 writes a source nasal-plus-velar cluster and this word has a single nasal.</td></tr>
<tr><td>烏珠留若鞮
Wuzhuliu Ruodi</td><td>ʔɑ tśo liu ńɑk te†</td><td>’a ço liu nyak te</td><td>Çal İnakt</td><td>the grey or dappled one, of the İnakt</td></tr>
<tr><td colspan="5">PHONETICS 烏 ʔɑ is vowel-initial, so the reading begins at 珠 tśo → ç, with 留 liu → l: an affricate written with the default tś- character and a liquid, both majority correspondences. Nothing is inserted, and no character is read as a consonant it does not carry.
REASON çal is “alaca, kır” in Kāşgarî: dappled, grey. Color epithets are a persistent Turkic naming pattern, as çakır “grey-eyed” shows in the same dictionary.</td></tr>
<tr><td>烏累若鞮
Wulei Ruodi</td><td>ʔɑ lui† ńɑk te†</td><td>’a lui nyak te</td><td>Uluğ İnakt</td><td>the Great, of the İnakt</td></tr>
</table>

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
PHONETICS 烏 *ʔɑ* → *u-*: a source syllable beginning with a vowel takes a Chinese glottal onset 137 times in 140, 97.9%, which is the strongest single correspondence used anywhere in this file. 累 *lui* → *-lu*: a source *l* is written with a Chinese *l-* 85 times in 88, 96.6%, and the vowel is rounded on both sides, 190 of 289, 65.7%. The final *-ğ* is not written, which the coda table of §8 puts at 8 of 21, 38.1%, the ordinary treatment of a source velar coda. Two characters, two syllables, nothing inserted and nothing discarded. The element costs 23.7%, about 1 in 4. **That figure is not comparable with a whole-name figure** such as the 1 in 5 for *Köngen* or the 1 in 37 that retires *Bayatur* in §8.1: it prices a two-syllable element, and a short string is cheap by construction, since every further character multiplies in another rate below 1. 若鞮 is not priced again here, being the one title element §9 argues once for all six rows that carry it. **REASON** *Uluğ* "great", from the Codex Cumanicus *Oulu*, glossed *magnus*. The word is among the most common elements in Turkic titulature, which is both the strength of this reading and its weakness: a very common word is an easy target to hit, and §6.3 of the paper measures choice by meaning as the weakest step in the whole procedure. **The same word was read at 呼都而尸道皋若鞮 in earlier drafts and is withdrawn there.** Step 54 measured what separates the two rows. This one has a glottal character for the vowel-initial syllable and a liquid character for the *-l-*; the long name has neither, and no character in it carries a liquid at all. Where one word appears twice in a record, the transcription and not the sense has to say which occurrence is real, and here it does. **What is not settled.** Whether the word is a title, an epithet or part of a personal name. The record gives no way to tell, and the other five 若鞮 rows are equally silent on it.				
呼都而尸道皋若鞮 Huduershi Daogao Ruodi	hɑ† tɑ ńə† śi douᴮ† kou ńɑk te†	ha ta nyı şi dou kou nyak te		
PHONETICS **The reading earlier drafts gave this name, Öz-Uluğ İnakt, is withdrawn, and none is proposed in its place.** It fails three measurements, none of them close. **First, twice over, on the zero this file uses elsewhere to retire a reading.** A source syllable beginning with a vowel takes a Chinese glottal onset 137 times in 140; it takes a velar 0 times in 140 and an *h-* 0 times in 140. Both *öz* and *uluğ* begin with a vowel, and the characters asked to write them are 呼 *hɑ* and 皋 *kou*. The same zero retired *Çöge-Ok* at 車牙若鞮. **Second, on the liquid.** A source *l-* is written with a Chinese *l-* 85 times in 88, 96.6%, and a source liquid onset of either kind, *l* or *r*, takes a Chinese *l-* 262 times in 288, 91.0%. Not one of the eight characters carries a liquid onset, so the *-l-* of *uluğ* would have to go unwritten in the one position the corpus almost always writes it. **Third, on the four characters left over.** Reading 呼 as *öz*, 皋 as *uluğ* and 若鞮 as the title leaves 都 *tɑ*, 而 *ńə*, 尸 *śi* and 道 *dou* carrying nothing. Across 1,017 pairs in the Sanskrit corpus no transcription uses more than one character beyond the syllables of its source: 772 are one to one, 152 use one extra, 93 use fewer, and the count of pairs with four or more spare characters is 0. Step 50 established that most of those single extras are a Chinese semantic suffix appended to a finished transcription, 鬼 鳥 國 城, which these four are not. **The reading cannot be priced, and that is the result.** A zero has no rate to multiply. **REASON** **No reading is proposed here.** This is the longest string in the record and the one where a reading is easiest to invent, because the longer the string the more ways there are to cut it into Turkic elements, and the cut then does the work that no individual correspondence is doing. That is the failure mode §6.3 of the paper measures, and it is what produced the withdrawn reading. *Uluğ* stays at 烏累若鞮, where the transcription supports it. **What the characters do carry.** Step 54 searched the lexicons at the classes the first six characters actually hold, 呼 a velar or *h*, 都 a dental stop, 而 a palatal nasal, 尸 a sibilant, 道 a dental stop, 皋 a velar. Every two-character frame returns hits and none of them reads as a ruler's name; they are listed in the step output as a starting point for someone else, not as candidates. **What is not settled.** What the name says. The Chinese sources gloss none of it, the string is long enough that the record cannot constrain a reading of it, and nothing here is evidence that it is or is not Turkic.				

TABLE S3 The recurring element itself, then the six later chanyu whose formal names end in it. Only the shared element is argued for in the paper; the material preceding it in each name is a proposal like any other in this supplement.

S1.8 The ruling clan name, in its two written forms

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
攣鞮 Luandi	lyan te†	lyan te	**El-indi**	state founder
PHONETICS 攣 *lyan* → *el* moves the liquid from the onset of the Chinese syllable to the coda of the Turkic one; no corpus shows that kind of metathesis. 鞮 *te* → *indi* expands one syllable into two. The reading is driven by the sense rather than by the transcription. **REASON** *el* "realm, state" with the definite past in *-di*: literally "he founded the realm". This is the clan name as the *Shiji* and *Hanshu* write it.				
虛連題 Xulianti	kʰɨɑ lianᴮ† de†	kıa lian de	**Kıyalığ El-indi**	state founder of the rocky region
PHONETICS 虛 *kʰɨɑ* → *kıya* is reachable. 連 *lianᴮ* → *lığ* adds an unwritten coda, and 題 *de* → *el-indi* carries a whole two-syllable word on one character. Three characters are being asked to yield five syllables. **REASON** The same clan name as 攣鞮, in the form the *Hou Hanshu* gives, with the *kaya* element recurring from 虛閭權渠.				

TABLE S4 攣鞮 in the *Shiji* and *Hanshu*, 虛連題 in the *Hou Hanshu*. Both are taken here as the same name, which is the usual view.

S1.9 Titles and lexical items

<table>
<thead>
<tr><th>FORM, AS WRITTEN IN THE HISTORIES</th><th>LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)</th><th>THE SAME READING IN TURKISH ORTHOGRAPHY</th><th>READING WE PROPOSE</th><th>SENSE WE PROPOSE FOR THAT READING</th></tr>
</thead>
<tbody>
<tr><td>單于
chanyu</td><td>dźan† wɑ</td><td>can va</td><td>Çarvuş</td><td>army commander; the earlier form of çavuş</td></tr>
<tr><td colspan="5">PHONETICS Two characters, 單 with a written coda and 于 open. 單 → çar- takes the second of the character’s two Later Han readings, the dental tɑn, for the onset: a source voiceless affricate written with the voiced dź- of the first reading is 0 of 48 in the Sanskrit corpus, while with a dental onset it is 8 of 48, 16.7%. For the coda it takes the Old Chinese value, Baxter and Sagart’s *dar, whose liquid becomes the -n of the later table; step 43 measures how often the two reconstructions differ in this way across the record. 于 wɑ → -vuş is the best value this character offers: a source v written with a Chinese w- character is 9 of 65, 13.8%, against 1 of 55 for a source b and 0 of 140 for a vowel-initial syllable. The final -ş is unwritten, which a sibilant coda is in 27 of 36 cases, 75%, since Later Han has no sibilant coda. The liquid is what makes the form possible. Across the 8,930 headwords of the Kāşgarî index and the Codex, -rv- and -rb- occur 18 times in native words (burba-, karva-, kurbaka, turbı), while -nv- occurs 0 times and -nb- 5 times, all in loanwords: a liquid before a labial is Turkic, a nasal before one is not. Two shapes the frame excludes outright, whatever word is tried. A velar first element is not available: a voiced affricate writing a velar stop appears once in 220 in the Sanskrit corpus and that one case is a translation mislabelled as a transcription, and characters that write kan exactly, 干 kɑn and 汗 gɑn^{c}, were available and not used. A vowel-initial second element is not available either: a glide character writing a vowel-initial source syllable is 0 of 140.
REASON Çarvuş, the earlier form of çavuş, “army commander”. A native Turkic title of the first rank. Clauson does not mark it foreign, unlike beg, xan and xağan, and Doerfer 1055 has it travelling outward, to Persian and thence to Urdu and Lezgian. It is attested in the eighth-century inscriptions in the highest possible company: bilgesi çavuşı erti, “he was his Counsellor and Army Commander” (Ix. 17), and at Tonyukuk 7 the speaker is Tonyukuk himself. Clauson glosses it “originally the principal military officer of a xağan, the military counterpart of bilge”, and adds that “as time went on the post gradually lost status”, ending as the modern Turkish sergeant; Kāşgarî defines the office as the one who orders the ranks in battle and restrains the troops from outrages (I 368), and his form of the word is çaβuş, with a bilabial fricative. Why it is proposed here. Founders of dynasties commonly bear military titles: Osman Gazi, Alp Arslan, Alp Tegin. And a supreme title that began as a military one is an ordinary trajectory, as with Latin imperator “commander” and dux. The Türk inscriptions are a thousand years later than the Xiongnu, so the decline Clauson describes was already under way when they were cut, and at the earlier stage the office Clauson calls the principal military officer of a xağan could have been the ruler’s own title rather than his officer’s. What is not settled. Çarvuş itself is not recorded in the later Turkic sources. Clauson’s Etymological Dictionary, Erdal’s Grammar of Old Turkic, Kāşgarî’s Dīwān, the Kutadgu Bilig, Doerfer’s TMEN and Tekin’s edition of the Irk Bitig all carry çavuş or çaβuş and none carries a form with the liquid. On this reading the -r- was lost before the eighth century, which is a thousand years after the name was written down.</td></tr>
<tr><td>撐犁孤塗單于
chengli gutu chanyu</td><td>ḍaŋ lei† kuɑ dɑ dźan† wɑ</td><td>dang lei kua da can va</td><td>(compound of the three rows below)</td><td></td></tr>
<tr><td colspan="5">PHONETICS Six characters forming the full ceremonial title, and the only item in the record the Chinese sources gloss in full. It decomposes into three units that each have their own row here: 撐犁 ḍaŋ lei, 孤塗 kuɑ dɑ and 單于 dźan wɑ. No separate reading is proposed for the compound; whatever the three parts are read as, the whole follows.
REASON The Hanshu glosses the phrase as “Son of Heaven, the Chanyu”, with 撐犁 = 天 “heaven” and 孤塗 = 子 “son”. That makes it the single most valuable item in the record, because both the form and the meaning are recorded, and it is the evidence on which most of the language-attribution argument has rested for a century.</td></tr>
<tr><td>撐犁
chengli</td><td>ḍaŋ lei†</td><td>dang lei</td><td>Tanglı</td><td>bearing divine light; sacred</td></tr>
<tr><td colspan="5">PHONETICS 撐 ḍaŋ → tang: retroflex ḍ for t is routine and the -ŋ coda is written. 犁 lei → lı: direct. Both steps are default. The same two syllables equally support tengri, which the Chinese gloss 天 and the lexicon search of §6.3 both favour: Kāşgarî returns it 4th, the Codex 2nd.
REASON tang of dawn or light, read as “bearing divine light; sacred”. Offered as an alternative to the standard identification with tengri, not as an improvement on it.</td></tr>
<tr><td>孤塗
gutu</td><td>kuɑ dɑ</td><td>kua da</td><td>Kutı</td><td>the blessing of heaven, with the third-person possessive</td></tr>
</tbody>
</table>

<table>
<tr><th>FORM, AS WRITTEN IN THE HISTORIES</th><th>LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)</th><th>THE SAME READING IN TURKISH ORTHOGRAPHY</th><th>READING WE PROPOSE</th><th>SENSE WE PROPOSE FOR THAT READING</th></tr>
<tr><td colspan="5">PHONETICS 孤 kuɑ → ku: the default velar onset, direct, with the coda open. 塗 dɑ → tı: a dental onset carrying the -t of kut, with the vowel free. Two characters, two syllables, nothing inserted and nothing discarded. The frame is k/g/h – t/d and both codas are unwritten, so a source -r, -l, -s or -z could hide in either. 66 attested headwords fit it.
REASON kutı, “the blessing of heaven”, with -ı as the third-person possessive. kut is the divine favour that legitimises a ruler: Clauson records it in the hendiadys kut kıv “divine favour”, and it is the word the Kutadgu Bilig is named from. The possessive is not padding but what Turkic requires of the second noun in a genitive-possessive phrase, tengri kutı, “the kut of heaven”, which is the shape of the full title 撐犁孤塗單于. The phrase is attested, not assembled. Clauson's entry for kut cites the Irk Bitig for teŋri kutınta “by the favour of heaven”, which is teŋri kutı with a locative on it, and the Orkhon inscriptions for the dative kutıŋa, in the passage where the ruler's mother is said to have enjoyed the favour of heaven. So both the possessive form and the two-word phrase are on record in Old Turkic, and the Chinese title maps onto them element for element. That is now checkable in the primary edition rather than only in Clauson’s report of it: Tekin’s glossary to the Irk Bitig carries kut as a headword, glossed “divine favor, blessing, good fortune”, and its citations include kutınta at omens 15 and 56, the first of them in the phrase teŋri kutınta. The manuscript is ninth-century. In Old Turkic that suffix is -ı / -i, so kutı rather than kutu; the -u of modern Turkish kutu comes from labial harmony, a later development. The two-character spelling is itself evidence for the second syllable. Had the word been a bare kut, a single character was available that writes it exactly, 骨 kuət, with the -t on the page, and the record uses that character in 骨都侯. A scribe who had the one-syllable option and wrote two characters was writing two syllables. The vowel of 塗 is ɑ and matches neither ı nor u closely, but §7 of the paper puts vowel recovery at 50.0% with the interval straddling one half, so the vowel is the least informative part of the spelling and cannot decide between them. The Chinese gloss is 子 “son”, and this reading does not translate it. That is recorded against the proposal rather than explained away. The word the gloss wants is oğul, and oğul is barred by the writing rather than merely disfavoured: it is vowel-initial while 孤 carries a velar, and in the Sanskrit corpus a source vowel-initial syllable took a zero or glottal onset in 96% of 183 cases and a velar in none of them. Its medial -ğ- has nowhere to sit either, since 塗 is dental. So oğul is not one of the 66 headwords that fit the frame and never was a candidate; the same wall excluded okluk at 胡祿. Of the 66 that do fit, none means “son”: they include kut “heavenly favour, fortune”, kat “layer”, kata “time, occasion”, ketü, Kerti verus and kötü. The Yeniseian rival, as its author now leaves it. Pulleyblank compared the word with Arin bĭkjàl “son”. In 2000 Vovin held that Arin form to be isolated in the family, the other languages reflecting Proto-Yeniseian *puʔ-b “son” (Ket hiʔp, Yug fiʔp, Kott fup), and found no Altaic etymology either (Vovin 2000, CAJ 44/1, p. 91). He withdrew it three years later: “I would like to take back my suggestion that Arin form is isolated in Yeniseian” (Vovin 2003, Part 2). He then rebuilds the comparison rather than dropping it, reading Pumpokol p-halla “my son” as a cluster with the first-person prefix rather than as an f-, which makes it cognate with the Arin form, and ruling out Pulleyblank’s Ket qal/qaloq instead. The Yeniseian side of this item is therefore stronger than Part 1 alone would suggest, and it is recorded here as he left it. So this is the one item in the record where two independent kinds of evidence point at different words: the writing supports kutı and the gloss asks for a word the writing excludes. The gloss can be accounted for without supposing the informant was confused about Xiongnu. If the title meant “the heaven-favoured chanyu”, the Chinese expression that fits it is 天子, “Son of Heaven”, which is their own imperial title. 天子 has two characters where the Xiongnu title has two words, so the element glosses follow mechanically from that equation: 撐犁 aligns with 天 and 孤塗 with 子. On that account the first gloss is a true lexical gloss, since 撐犁 is teŋri “heaven”, and the second is a back-formation from the phrase-level equivalence rather than testimony about what 孤塗 means. This is an explanation of the gloss and not a test of it, which is why the reading is offered here rather than argued in the paper.</td></tr>
<tr><td>屠耆
tuqi</td><td>dɑ gɨ†</td><td>da gı</td><td>Doğru</td><td>straight, upright, honest</td></tr>
<tr><td colspan="5">PHONETICS 屠 dɑ → do: a dental onset, direct. 耆 gɨ → ğru: a velar onset, with the -r unwritten, which is forced rather than permitted, since Later Han carries no -r in any coda position. 144 attested headwords fit the frame, so the transcription does little to narrow the choice; the vowels are free as always.
REASON This is the author's proposal, and the comparison is ours. We have not found this reading proposed in print. The search that would have found it covered every book held for this project, including Golden's Introduction to the History of the Turkic Peoples in a machine-readable form made for the purpose, and returned nothing. It is recorded here as our own comparison and not as a reading taken from the literature. The Codex Cumanicus has Togru glossed rectus. The Chinese sources gloss 屠耆 as 賢 “worthy, wise”, and it appears in 左右屠耆王, the Wise Kings of the Left and Right. The senses meet more closely than “straight” suggests. Clauson derives toğuru: / toğru: as a gerund of toğur-, physically “straight” and then metaphorically “straight, honest, upright, true”, and that moral sense is old rather than modern: a fourteenth-century glossary renders it with Arabic al-tiqa “trustworthy, honest”, and toğru ayt- with ṣadaqa “to tell the truth”. The word survives across the whole Turkic family and as modern Turkish doğru, “straight, correct”. Toğrıl and Toğrul are not offered in support. The two names resemble toğru in sound, and they are well attested: Kāşgarî records that men are called Toğrıl, an Alp Toğrul Tégin appears in a Uyghur list of proper names, and the first Seljuk sultan bore the name. But Toğrıl is a bird of prey, and Clauson derives it from toğra- rather than from toğur-, noting that it never survived as a common noun. What makes the two look like one word is a separate homophone, the passive toğrıl- “to be upright”, which is from toğur- and survives as modern Turkish doğrul-. So the names are recorded here to head that resemblance off, not to rest anything on it: the case for Doğru stands on the Chinese gloss 賢 and on the attested moral sense of toğru. The paper does not argue this reading; §9.1 there tests only proposals taken from the literature.</td></tr>
<tr><td>谷蠡
luli / guli</td><td>kok† le^{B}†</td><td>kok le</td><td>Körklü</td><td>glorious, imposing</td></tr>
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<tr><td colspan="5">PHONETICS 谷 kok → körk: the default velar onset, with the -k coda written and the -r- unwritten before it. Later Han has no -r coda, so a source -rk cannot be written at all without spending a further character on an l- syllable; leaving it unwritten is the only option two characters allow, and it is the majority practice: step 31 counts 63 verified pairs whose source form has a liquid before a consonant, and the liquid is unwritten in 54 of them, 86% (達磨 for dharma, 薩達摩 for saddharma, 播鉢吒 for parpaṭa). In 6 of the 63, 10%, the scribe spent a further character carrying an l- onset instead, so the step this reading asks for has a measured minority against it rather than none. 蠡 le^B → lü: the l- onset direct, with an open coda. Two characters, two syllables, one unwritten consonant. Both are polyphonic characters: the table gives 谷 as kok, jok and lok, and 蠡 as le^B and loi^B; this reading takes the first of each.
REASON Körklü, “glorious, imposing”. The word is a headword in both lexicons used here: Kāşgarî has körklüg, glossed iyi, güzel ve gösterişli, “good, beautiful and imposing”, and the Codex Cumanicus has körk glossed decus, glory or dignity rather than prettiness. Clauson records the form across the whole span, with Kipchak and Coman körklü / körkli and Osmanlı görklü. The Kipchak form matters for the spelling: körklü carries no final -g, so the open coda of 蠡 is simply correct and nothing further needs licensing. The word is used of exalted figures and not only of pretty objects: an Old Turkic Buddhist text has teŋridem kız azu teŋri oğlanı teg körkle oğlan, “a child as beautiful as a divine girl or a son of God” (TT V 12, 127, via Clauson). Two caveats belong with that citation: Clauson calls the text a fairly late one, and he queries whether the -le form körkle / körtle is native rather than borrowed. The form this reading uses, körklüg, is not in doubt, being a straightforward derivation in -lüg from the attested körk. An authority assigns 谷 a different reading in this very title. Pulleyblank's Lexicon, at K1202a, glosses 谷 read lu as “luli king, Xiongnu title”, cross-referring gu and yu as the other readings. That is the lok branch, on which no Turkic word can fill the first syllable at all, so if Pulleyblank is right this reading is not available and 谷蠡 is not a Turkic word. The proposal is recorded with that against it, unresolved. The other readings of 谷 do not compete. On lok, which the office's conventional name, the Luli Kings, would suggest, the first syllable begins with a Chinese l-, and a Chinese l- writes a source l- or r-. Turkic admits neither initially: Clauson calls initial r-, like initial l-, a sound entirely foreign to the language, and loanwords that established themselves took a prosthetic vowel. That branch yields no candidates at all, so if 谷 is lok here, 谷蠡 is not a Turkic word. On jok, which is Schuessler's palatal glide, the frame is y…k plus l/r and admits yüklüg “laden” and yokla- “to rise, ascend”, neither of which suits an office. What cannot be tested here. The Chinese sources gloss 撐犁 and 孤塗 and say nothing about 谷蠡, so this row carries no semantic constraint at all, only the rank of the office: the 谷蠡王 of the Left and Right stood among the four highest, paired with the Wise Kings. The choice among the candidates therefore rests on the epithet argument that §S1.3 names as this supplement's main weakness, and the reading is offered as permitted rather than confirmed.</td></tr>
<tr><td>當戶
danghu</td><td>tɑŋ† ɡɑ^B</td><td>tang ga</td><td>Tamga</td><td>the ruler’s seal, and the officer who keeps it</td></tr>
<tr><td colspan="5">PHONETICS 當 tɑŋ → tam-: a source t written with a Chinese t- is 55 of 198, 27.8%, and a source a written with an open vowel is 1045 of 1509, 69.3%. The coda is the weak link and is stated rather than buried: the word carries an -m and the character carries the velar -ŋ, a change of place which the corpus permits in 15 of 53 cases, 28.3%, but does not favour. 戶 ɡɑ^B → -ga: a source g written with a Chinese g- is 52 of 90, 57.8%, with the vowel open again. The pair is the corpus device for a medial nasal-plus-velar: given a written -ŋ coda followed by a velar onset, the source had a nasal followed by a velar stop in 26 of 29 cases, and the three exceptions are all one word. The whole reading costs 2.18%, about 1 in 46. That is more expensive than the 1 in 37 which retires Bayatur in §8.1 of the paper, and the figure is given here so that a reader can weigh it rather than take the reading on trust. What these characters cannot write. A word with a single velar nasal and no velar stop, which is why toŋa stood here in earlier drafts and has been moved to 車牙若鞮, where two characters write it directly. The Codex Cumanicus forms that appear to fit this frame, tengis, tongus, tangis, are medieval Latin spellings of a single velar nasal, as the Dīwān’s own teñiz and toñuz show for the same words, and they are excluded for the same reason.
REASON Tamga, the seal of the ruler, and by extension the officer who keeps it. 當戶 is an administrative office in the Xiongnu hierarchy rather than a ruler’s name, so the register wanted is court vocabulary, and tamğa is exactly that. The sense is attested, though not under the headword. Kāşgarî’s own entry for tamga gives an unrelated sense, a branch of water or an anchorage, but the derivatives in the same dictionary carry the seal throughout: tamgalamak, “to strike the khan’s seal” (III 353), and tamgalıg and tamgalık, “objects bearing the khan’s seal” (I 527). Clauson adds an Uyghur civil document, beg tamğası elgiŋde, “a beg’s seal is in your hand” (TT I 129). So the administrative sense is on record in two independent places, and the later Turkic and Mongol chanceries make tamga the ordinary word for the seal, with tamgacı its keeper. What is not settled. The price, 1 in 46, which is worse than the threshold the paper uses elsewhere, and the reader is told so above. The nasal, which is a change of place rather than a direct correspondence. That the seal sense reaches the headword only through its derivatives, so the chain has one link more than at a row where the dictionary glosses the word itself. And whether the Chinese office 當戶 had anything to do with seals, which the histories do not say and this reading assumes.</td></tr>
<tr><td>閼氏
yanzhi / ezhi</td><td>ʔɑt dźe^B†</td><td>’at ce</td><td>Elti</td><td>lady; a woman related by marriage</td></tr>
</table>

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<tr><td colspan="5">PHONETICS 閼 ʔɑt → el-: a vowel-initial character whose written -t carries the source liquid. Later Han had no -r or -l in any coda, so a source syllable ending in one was written with a -t or left open: of 37 such syllables in the Sanskrit corpus, 21 were left open and 14 took a -t, 38%. 氏 dźe^{B} → -ti: a dental onset with the vowel free. The first character is not polyphonic: Schuessler's table gives 閼 the single value ʔɑt, and Baxter and Sagart give Middle Chinese ʔat from Old Chinese *qˤat. The -n that the parallel spelling 焉支 carries belongs to 焉, a different character, which Baxter and Sagart give two readings, both in -n. The second character is another matter: the table gives 氏 as dźe^{B}, gie^{B}, tśe and kie, and Baxter and Sagart reconstruct *k.deʔ with a velar preinitial, which is where the velar readings come from. This item is character-anchored in the emission row; the second character is less constrained than that alone suggests.
REASON Elti, "lady". Modern Turkish elti is what each of two brothers' wives calls the other, and in dialect what two wives of one man call each other. It is not only modern: Clauson records élti: glossed "lady", standing opposite al-jāriya "slave girl", in a fourteenth-century Arabic-Turkish glossary, which is the sense a consort's title would want. Nişanyan takes Middle Turkic élti "married woman, as a respectful term" and derives it, with a stated doubt, from Old Turkic élt- "to carry, to lead". If that holds, the word has the shape of kelin "bride", which Clauson derives from kel- as "one who comes in (to the family)": a deverbal noun naming a woman by how she entered it. 閼氏 is the title of the chanyu's principal consort. Why this replaces the earlier proposal in this supplement. The earlier reading was ece "elder sister", later eci "senior woman, lady", both of which are in the Kāşgarî index (I, 86 and I, 87) and the second of which is the better semantic fit. Both discard the written -t of 閼, and the corpus makes that expensive: where the source has a single affricate or sibilant between two vowels, the preceding character was open in 134 of 143 junctions, 93.7%, and carried a -t in 2, 1.4%. That is a worse cost than the 1 in 37 the paper uses in §8.1 against a rival reading elsewhere, so consistency required either conceding it or finding a reading that uses the -t. elti uses it, at 38%. The earlier proposal is recorded here rather than removed, since the reasoning that led away from it is the useful part. What is not settled. elti is not in the Türk Dil Kurumu index nor in the Codex Cumanicus, and the fourteenth-century attestation is in the edition Clauson marks "(only)". Pulleyblank read the title as Turkic and Mongolian qatun; Vovin rejects that and proposes instead a Yeniseian *alte or *elte "wife", from Proto-Yeniseian *ʔal(V)it- (Kott alit, Assan alit, Arin biqam-alte), noting that it "could be transcribed in Chinese only as *ʔat-tej or *ʔen-tej". He is right about the Chinese and the same spelling writes elti by the same steps, so the transcription does not choose between the two languages. §9.1 of the paper sets that out as a measured case of the negative result of §6.1.</td></tr>
<tr><td>骨都侯
guduhou</td><td>kuət tɑ go</td><td>kuıt ta go</td><td>Kutak</td><td>the blessed one</td></tr>
<tr><td colspan="5">PHONETICS 骨 kuət → kut: default velar onset with the -t coda written, the tightest single syllable in this supplement. 都 tɑ then begins with the same dental, and Table 8 of the paper measures what that means: at a junction where a written coda is followed by the same-place onset, the source word carries that consonant once rather than twice in 121 of 140 cases (86%). So 骨都 transcribes kuta, not kutta. 侯 go → -k closes it, a Later Han voiced g- standing for a source k more often than for a source g.
REASON kut is "kut, uğur, devlet, baht, talih, saadet" in Kāşgarî (heavenly favour, fortune, sovereign grace) and Clauson records the verb kutad- "to be blessed", the stem behind the Kutadgu Bilig. Kutak itself is not attested: it would be a deverbal or diminutive formation in -(A)k on that stem, which is a live suffix but not one witnessed on this word. Compare Qutalmış, ancestor of the Seljuks of Rum. Note also that kut is the Turkic candidate for 孤塗, where the Chinese gloss 子 "son" fits the Yeniseian proposal better and not this one.</td></tr>
<tr><td>且渠
juqu</td><td>tsɑ^{B}† gɨɑ</td><td>tsa gıa</td><td>Çaka</td><td>the striker; from çak- 'to strike fire, to strike a blow'</td></tr>
<tr><td colspan="5">PHONETICS 且 tsɑ^{B} → ça: ts- for a source affricate is a minority spelling but an attested one, 15% in the Sanskrit corpus and 9% in Ligeti, against a tś-character as the default at 37%. 渠 gɨɑ → ka: a Later Han voiced g- stands for a foreign k more often than for a foreign g. Both steps are permitted by the measured correspondences.
REASON çak- "to strike fire from flint, to sting, to slander", which Clauson describes as onomatopoeic in character and of violent action, physical or mental. No çaka headword exists in Kāşgarî, the Codex or Clauson, only the verb and its derivatives çakmak, çakır, çakılmak, so Çaka would have to be the -A converb of that stem, and a bare converb serving as an office title is not a demonstrable pattern. The 11th-century emir Çaka Bey of Smyrna (Byzantine Τζαχᾶς, fl. 1081–93) shows that çak- could yield a personal name, but that is a personal name set against a title. The Chinese sources do not record what the office of 且渠 did, so no sense can be tested against a gloss.</td></tr>
<tr><td>徑路
jinglu</td><td>keŋ^{C} lɑ^{C}</td><td>keng la</td><td>Kıngrak</td><td>a large cleaver-like knife</td></tr>
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<tr><th>FORM, AS WRITTEN IN THE HISTORIES</th><th>LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)</th><th>THE SAME READING IN TURKISH ORTHOGRAPHY</th><th>READING WE PROPOSE</th><th>SENSE WE PROPOSE FOR THAT READING</th></tr>
<tr><td colspan="5">PHONETICS 徑 keŋ^{c}</i> → <i>kıng</i>: default velar onset with the -ŋ coda written. 路 <i>lɑ^{c} → ra: a Chinese l- for a source r-, the direction those counts show in 12 of 14 attestations. The Turkic word's final -k is not written, which Table 7 of the paper licenses at 30%.
REASON Not our reading. Friedrich Hirth compared 徑路 with Turkic kıŋrak in The Ancient History of China to the End of the Chóu Dynasty (1908), pp. 65–70, calling it the oldest Turkish word on record. Kāşgarî glosses kıñrak as a knife like a butcher's cleaver, for cutting meat and dough, and the Chinese sources describe 徑路 as a blade used in oath-taking, so gloss and referent agree across unconnected sources. Turkic origin, on Clauson's analysis. Clauson derives kıŋra:k as a deverbal noun from an unattested *kıŋra:-, denominal from kıñır, in the sense of something curved or something that cuts crookedly, so the word is motivated inside Turkic by the standard authority rather than by us. Kāşgarî's index carries the same family (kıñru “askew”, kañrak “palate”, koñur- “to wrench, bend back”), and the Codex glosses Kingir as curvus. Clauson also records the word alive in Teleut as kıŋırak and Kirghiz kıŋarak / kıŋırak, a rough two-edged knife for cutting felt and scraping hides. Two cautions. Kāşgarî's text at III 382 reads kırpa:k, corrected in the margin to kıŋra:k, so the form rests on an emendation; and of the family it is kıñır that fits the transcription exactly, while kıñrak, the member meaning a knife, needs the unwritten final -k allowed at 30%. Iranian origin. The object has also been derived from an Iranian word for a short sword, the akinakes type, historically plausible, since the Xiongnu's western neighbors were Iranian-speaking. But 路 is a liquid-onset character, so the channel requires a liquid in second position, and the akinakes frame (k–n–k) has none. As stated, that account fails on the frame. Two cautions there too: we have not seen the exact Iranian etymon its proponents advance, and the channel shows what a transcription permits, never what it requires. §9.1 of the paper sets both origins out.</td></tr>
<tr><td>服匿
funi</td><td>buk† ṇɨk</td><td>buk nık</td><td>Buknik</td><td>clay jar (küp)</td></tr>
<tr><td colspan="5">PHONETICS 服 buk → buk: exact, including the written -k coda. 匿 ṇɨk → nik: retroflex ṇ for n is routine and the second -k is also written. This is the closest fit to its own transcription of anything in this supplement: no segment inserted, none discarded.
REASON A clay jar (Turkish küp). One of the few Xiongnu material terms the Chinese sources preserve, which is what makes it worth a reading at all.</td></tr>
<tr><td>甌脫
outuo</td><td>ʔo t^{h}uɑt</td><td>ʼo tuat</td><td>Otuyat</td><td>the border watchtowers</td></tr>
<tr><td colspan="5">PHONETICS 甌 ʔo → o: vowel-initial, direct. 脫 t^{h}uɑt → tuyat: the aspirated t- is direct and the -t coda is written; the only addition is the medial -y-.
REASON The border watchtowers, or the empty zone between Xiongnu and Han territory, for which the Chinese sources use 甌脫.</td></tr>
<tr><td>胡祿
hulu</td><td>gɔ† lok</td><td>go lok</td><td>Külüg</td><td>famous, renowned</td></tr>
<tr><td colspan="5">PHONETICS 胡 gɔ → kü: the devoicing is normal, a Later Han voiced g- standing for a source k more often than for a source g; ü cannot be written and surfaces as o. 祿 lok → lüg: the -k coda is written. The consonant frame k–l–k is secure and only the vowels are free.
REASON külüg “famous, renowned” is an Old Turkic epithet of praise, recorded by Clauson; note that the form is külüg, not külük, and it is not a headword in either lexicon used here. Two objections stand. First, the Chinese-side record gives the meaning of 胡祿 as “quiver”, not a term of praise, though that gloss is itself marked low-confidence and chronologically shaky. Second, the reading the gloss wants is barred by the transcription: the one Turkic quiver word whose skeleton fits is okluk (Kāşgarî, glossed sadak), and it is vowel-initial, whereas 胡 carries a voiced velar. In the Sanskrit corpus a source vowel-initial syllable was written with a zero or glottal onset in 96% of 183 cases, and with a velar in none. So this is an item where the Chinese gloss names a target, a Turkic word exists for it, and the spelling rules that word out, recorded because a negative of that shape is worth more than a resemblance.</td></tr>
</table>

TABLE S5 The titles and the handful of common nouns the Chinese histories preserve, in the order of Appendix A. The lexical items are the more interesting group because some of them are glossed in the source, which means a proposal about them can in principle be checked.

S1.10 Ethnonyms

<table>
<tr><th>FORM, AS WRITTEN IN THE HISTORIES</th><th>LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)</th><th>THE SAME READING IN TURKISH ORTHOGRAPHY</th><th>READING WE PROPOSE</th><th>SENSE WE PROPOSE FOR THAT READING</th></tr>
<tr><td>匈奴
Xiongnu</td><td>huoŋ nɑ</td><td>huong na</td><td>Hung</td><td>the ethnonym, from the first character alone</td></tr>
<tr><td colspan="5">PHONETICS 匈 huoŋ → hung: direct, with the -ŋ coda written. 奴 nɑ is discarded rather than read, on the assumption that 奴 “slave” is a Chinese pejorative appended to the name rather than part of it.
REASON Dropping 奴 is common in the literature and what is proposed is only the vocalisation of the remaining syllable.</td></tr>
<tr><td>羯
Jie</td><td>kɨat</td><td>kıat</td><td></td><td>no reading proposed; kıyat withdrawn</td></tr>
</table>

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
PHONETICS A single character, *kiat*, with the *k-* default and the *-t* coda written, so the consonant frame *k…t* is secure. **The vowel carries information, and step 53 counts it.** Characters whose Schuessler vowel contains *ɨa* write a source *a* in 28 of 28 cases; a source *e* is 2.0% of the corpus and appears in none of them. On the Later Han layer this character writes a source *a*. **That measurement is recorded here and deliberately not used as an argument.** The Jie appear in the fourth century CE and the corpus behind the figure is Later Han, which is the period mixing §7 of the paper warns against; and Proto-Yeniseian **keʔt* is a reconstruction of a proto-form, so it does not fix the vowel of the fourth-century word and the measurement is not aimed at what Vovin actually claims. The one-syllable words the frame admits are *kat* "layer" and *kad*, which Kāşgarî glosses "kar fırtınası, insan öldüren bora, tipi", a blizzard or a killing gale. Neither is an obvious ethnonym.				
REASON **No reading is proposed here.** An earlier draft read this as *kıyat*, "of flooding". That is withdrawn, and the reason is not a fine judgement: *kıyat* is not a headword in the Kāşgarî index, the Codex or the *Irk Bitig* glossary, and the root it would be built on means the opposite of what was claimed. Kāşgarî's *kıy-* family is *kıymak* "sözden dönmek; kıymak, eğrilemesine doğramak", to break one's word and to cut crosswise, with *kıyık* "cayma, caymak" and *kıyturmak* "iğrilemesine kestirmek". Nothing in it is about water. The sense offered had no source. **The reading in print stands unopposed.** Pulleyblank and Vovin read the ethnonym as **ket* and connect it with Proto-Yeniseian **keʔt* "person, human being", the Yeniseian self-designation, reflected in Ket *keʔt*, Yug *keʔt*, Kott *hit*, Arin *kit/qit* and Pumpokol *kit* (Vovin 2000, *CAJ* 44/1, p. 91). This is the only item in the record where a reading from the literature stands with nothing of ours beside it, and it is left that way. **Why the vowel count is not used against it.** It would tell against **ket*, and this author has an interest in its telling against **ket*. That is the reason to state the two objections above rather than the conclusion, and to leave the row to the reading that has a proposer and a family of cognates behind it.				

TABLE S6 Group names rather than personal names, and subject to a further difficulty: a Chinese exonym need not transcribe anything the group called itself.

S1.11 Characters with more than one Later Han reading

CHARACTER	READING PRINTED	OTHER READINGS IN THE TABLE	ITEMS IN WHICH IT APPEARS
鞮	te	de	且鞮侯 壺衍鞮 握衍朐鞮 若鞮 復株累若鞮 搜諧若鞮 車牙若鞮 烏珠留若鞮 烏累若鞮 呼都而尸道皋若鞮 攣鞮
犁	lei	li	呴犁湖 撐犁孤塗單于 撐犁
且	tsɑ^{B}	tsʰia^{B} · tsʰiɑ · tsiɑ	且鞮侯 且渠
呼	hɑ	hɑ^{C}	呼韓邪 呼都而尸道皋若鞮
單	dźan	tɑn	單于 撐犁孤塗單于
累	lui	lyai^{B} · lyai^{C} · luɑi^{B}	復株累若鞮 烏累若鞮
衍	jan^{B}	jan^{C}	壺衍鞮 握衍朐鞮
冒	mək	mou^{C}	冒頓
呴	huo	huo^{C}	呴犁湖
復	buk	bu^{C}	復株累若鞮
搜	so	ṣu	搜諧若鞮
支	tśe	kie	郅支
曼	mɑn^{C}	mɑn · muɑn^{C}	頭曼
服	buk	bu^{B}	服匿
氏	dźe^{B}	gie^{B} · tśe · kie	閼氏
當	tɑŋ	tɑŋ^{C}	當戶
稽	kʰei^{B}	kei	稽粥
耆	gɨ	tśi^{B} · ki^{B}	屠耆
而	ńə	ńɨ	呼都而尸道皋若鞮
胡	gɔ	gɑ	胡祿
臣	dźin	gin	軍臣

CHARACTER	READING PRINTED	OTHER READINGS IN THE TABLE	ITEMS IN WHICH IT APPEARS
蠡	leᴮ	loiᴮ	谷蠡
谷	kok	jok · lok	谷蠡
車	tśʰa	kɨɑ	車牙若鞮
連	lianᴮ	lian	虛連題
道	douᴮ	douᶜ	呼都而尸道皋若鞮
邪	ja	ziɑ · zia	呼韓邪
頓	tuənᶜ	tuət	冒頓
題	de	deᶜ	虛連題

TABLE S7 Every character in the record for which Schuessler's table gives more than one Later Han reading, with the reading printed in Appendix A of the paper and the alternatives not used. Generated from `data/derived/polyphony.csv`, which the numbered script step 24 writes, so this table cannot drift from the repository. The daggers in the tables above mark these same polyphonic characters. Where an alternative would change a proposed reading, the entry for that item says so.

S1.12 The two reconstructions of each character compared

CHARACTER	LATER HAN (SCHUESSLER)	OLD CHINESE (BAXTER & SAGART)	ONSET CLASS	CODA CLASS
粥	tśuk	t-quk	palatal → velar	—
軍	kun	[k]ʷəər	—	dental nasal → liquid
臣	dźin	[g]i[ŋ]	palatal → velar	dental nasal → velar nasal
稚	ḍi	lrəəj-s	dental → liquid	—
維	wi	ɢʷij	glide → velar	—
詹	tśam	[t]am	palatal → dental	—
衍	jan	N-q(r)anʔ	palatal → velar	—
權	gyan	[g]ʷrar	—	dental nasal → liquid
韓	gɑn	[g]ˤar	—	dental nasal → liquid
邪	ja	səə.ɢA	palatal → velar	—
支	tśe	ke	palatal → velar	—
若	ńɑk	nak	palatal → dental	—
車	tśʰa	[t.qʰ](r)A	palatal → velar	—
而	ńə	[g]ew	palatal → velar	—
尸	śi	l̥[əə]j	palatal → liquid	—
道	dou	lˤuʔ-s	dental → liquid	—
單	dźan	dar	palatal → dental	dental nasal → liquid
于	wɑ	ɢʷ(r)a	glide → velar	—
塗	dɑ	lˤa	dental → liquid	—
閼	ʔɑt	qˤat	glottal → velar	—
氏	dźe	k.deʔ	palatal → dental	—

CHARACTER	LATER HAN (SCHUESSLER)	OLD CHINESE (BAXTER & SAGART)	ONSET CLASS	CODA CLASS
路	lɑ	Cəə.rˤak-s	—	open → velar stop

TABLE S8 The 22 characters of the record, of the 60 present in both reconstructions, where the two give a different class of segment in at least one position; the other 38 agree. This is the measurement §11 of the paper reports, at 37%. It is not a correction to Appendix A: Baxter and Sagart reconstruct a stage several centuries before the transcriptions and Schuessler one several centuries after, so the two bracket the moment of writing rather than dating it, and this table is the width of that bracket. Their **-s* and **-ʔ* are treated as having become tones by the Han and are not counted as codas, and a trailing **-j* or **-w* is read as the offglide of a diphthong rather than as a consonant. Generated from `data/derived/period_depth.csv`, which the numbered script step 43 writes.

S1.13 How to disagree

The repository includes `src/lookup_lexicons.py`, which searches Kāşgarî's *Dīwān*, the *Codex Cumanicus* and any EPUB lexicon placed beside them, folding the spelling conventions that separate Turcological from Turkish orthography (q ≡ k, ñ ≡ ng, ş ≡ s, ı ≡ i). Type any proposed form into it and see what the medieval sources actually have. That is how the *Codex* attestation for *inak* was found, and how the `-tI` count that ruled out the competing suffix was made.

For the reader who disagrees with our proposed names, we suggest disagreeing not with "this is unlikely", since most of them are, but with "this word is not attested before date X", or "this suffix does not attach to that stem", or "the Chinese character cannot carry that coda". Those are the objections that would be useful for the author.

S1.14 Status

Thirty-seven of the forty readings in Tables S2 to S6 carry the same label in the machine-readable file: *author's proposal; the paper measures the steps it takes for the name but does not present it as a finding*. That label is the honest summary of them, and it is not weakened by the fact that some are more attractive than others. Three rows carry a different status, and §S1.1 names them: **若鞮** as *İnakt*, which the paper argues and which meets its standard; **徑路** as *Kıngrak*, a comparison taken from the literature rather than made here, which the paper then tests; and one compound that is read through its constituent rows.

Twelve rows changed status while this supplement was being written, and the changes are recorded here rather than quietly absorbed. **孤塗** and **谷蠡** had no proposal at all in earlier drafts and now carry one. **冒頓** was argued in the paper in an earlier draft and is now an author's proposal like the others: §9.2 of the paper prices the three values its second character can take and does not choose among them, and the proposal made here is set out in S1.4 with its objections. **屠耆** was carried as a comparison taken from the literature and tested in the paper, and is now an author's proposal like the others: the reading could not be traced to any published source, and the paper no longer argues it. **閼氏**, **單于** and **軍臣** carry different readings from the ones earlier drafts gave them, in each case because a measurement made since told against the earlier reading; the entries state what the measurement was. At **軍臣** it was that a source voiceless affricate written with a voiced `dź-` is 0 of 48, which retired *Künçin* and sent the reading to the second of the character's two tabulated values. At **狐鹿姑** the reading itself is unchanged, but two objections the entry raised against it are withdrawn: leaving a liquid unwritten before an open character is the majority spelling and not an impossibility, and a written coda reaching forward to the next onset is the commonest use of such a coda. Both had been judged by ear before they were counted. At **揭** the proposal was withdrawn outright and none put in its place: *kıyat* is not a headword anywhere and the root it needed means to cut and to break one's word, so the sense offered had no source. That row now leaves the field to the reading in print, and §S1 records why a measurement that would have told against that reading is reported there but not used. At 呼都而尸道皋若鞮 the proposal was withdrawn too, and for measurements rather than for want of a source: a source vowel-initial syllable takes a Chinese glottal onset 137 of 140 and a velar 0 of 140, and both elements of the reading sat on velar characters; no character in the name carries a liquid, where a source l- takes a Chinese l- 262 of 288; and the reading left four of the eight characters carrying nothing, where no pair in the 1,017-pair Sanskrit corpus uses more than one character beyond its source syllables. *Uluğ* keeps its place at 烏累若鞮, which has a glottal character for the vowel-initial syllable and a liquid character for the *-l-*. That is the point of keeping the file: the same word was proposed in two rows, and the transcription, not the sense, said which one could hold it. At **車牙若鞮** it was that a source vowel-initial syllable written with a Chinese `ŋ-` is 0 of 140, which retired *Çöge-Ok*; the reading that replaces it, *Tunga*, was carried at **當戶** in earlier drafts and is moved here because those characters write a nasal followed by a velar stop and this word has a single nasal. **當戶** carries *Tamga* in its place, with its price stated in the entry.

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